# Speech to Emotion (STE)

## Abstract

Speech to Emotion (STE) is an application that detects emotions from a person's speech. Speech emotion recognition systems play an important role in improving human to computer communication in NLP applications. Speech features were extracted with MFCC and the prediction performance of SVM, MLP and LSTM models was compared. The LSTM model achieved the highest performance, with an Accuracy, Recall, Precision and F1-score of 88%. This LSTM model was utilized by the STE application to detect emotions from either live recordings or pre-recorded audio samples. Due to the variability in human speech, further enhancements are needed to improve the predictive performance of the system in real-world scenario


## 1. Introduction and Background

Emotion is a fundamental part of human language and recognizing emotion from a person's speech is essential for improving language understanding in Natural Language Processing (NLP) applications. The advancement of Artificial Intelligence (AI) techniques has led to the possibility of developing Speech Emotion Recognition (SER) systems, enabling machines to recognize the emotional state of a person based on their speech. These SER systems are useful across a wide range of domains. Hence, SER systems are a highly sought out tool especially with the increased popularity of AI chat bot systems.

The objective of this project was to create a SER system by utilizing AI tools. The STE application have the features of detecting emotion from either a live audio recording or from an imported audio sample. This application can be significant as it helps improve communication between person-to-person and machine-to-person. Furthermore, it can rapidly analyse the emotion of lengthy speech data. Thus, the STE application provides the users with a tool that can aid with their assessment of monitoring a person's emotional condition.

## 2. Methodology

### *2.1 Problem domain – Steps of methodology*

The creation of the SER system involves obtaining the speech input from the user from either a microphone recording or an imported audio sample. The audio data was pre-processed before the features were extracted. The extracted features were scaled before model training and testing were applied. The application presents the user with the text output and the emotion, predicted by the best performing AI model.

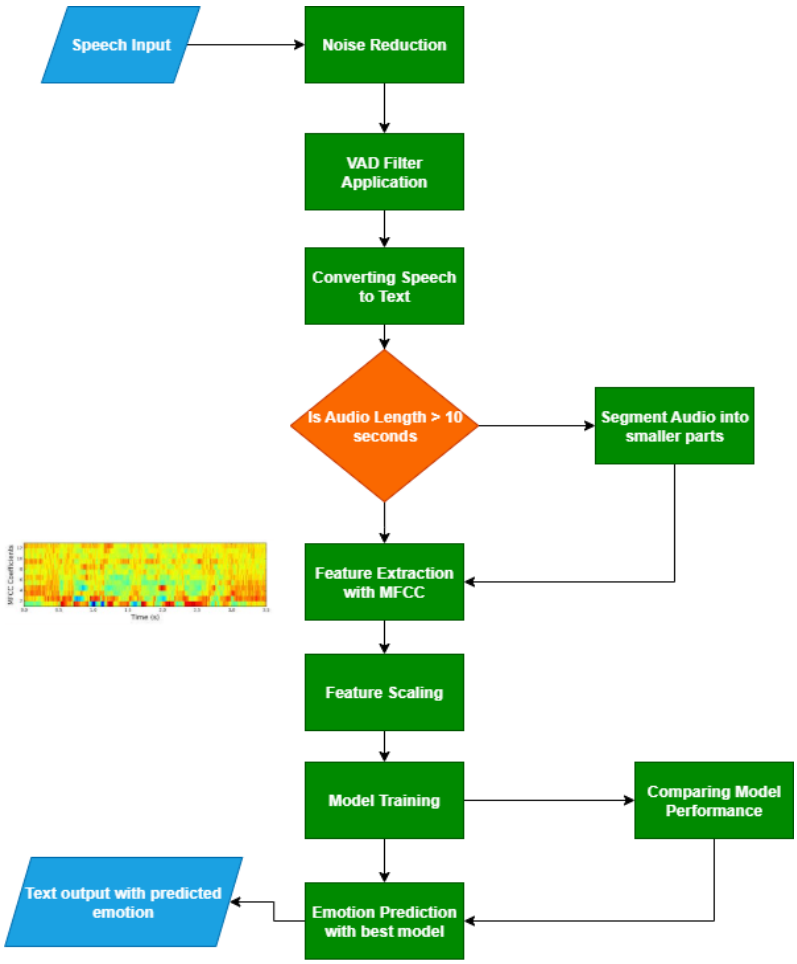



**Figure 1.** *System Workflow of the STE application*

*2.2 Dataset*

Predicting emotion from speech requires a large variety of data due to variations in human speech. To improve model generalizability, three datasets were used for training and testing:

1. RAVDESS - Ryerson Audio-Visual Database of Emotional Speech and Song
2. SAVEE – Surrey Audio-Visual Expressed Emotion
3. TESS – Toronto Emotional Speech Set

Each dataset contains short audio clips of actors reciting short phrases, expressed in seven different emotions: angry, sad, neutral, happy, surprised, fearful, disgust

| | Sample Size | Actors | Sample Length |
|---|---|---|---|
| **RAVDESS** | 1440 | ➢ 24 actors<br>➢ 12 male and 12 female<br>➢ Neutral North American accent | ~ 3 seconds |
| **SAVEE** | 480 | ➢ 4 actors<br>➢ All male<br>➢ British accent<br>➢ Aged 27-31 | ~ 3-5 seconds |
| **TES** | 2800 | ➢ 2 actors<br>➢ Both female actors<br>➢ 1 young and 1 old actor | ~ 2 seconds |

**Table 1.** *Summary information about the RAVDESS, SAVEE and TES datasets including sample size, actors and the average sample length of the audio data*

*2.3 Exploratory Data Analysis*

The overall distribution of samples in the dataset is balanced, with each of the seven emotion classes containing more than 600 samples (Figure 2). This balanced distribution minimizes class bias in the model's prediction.

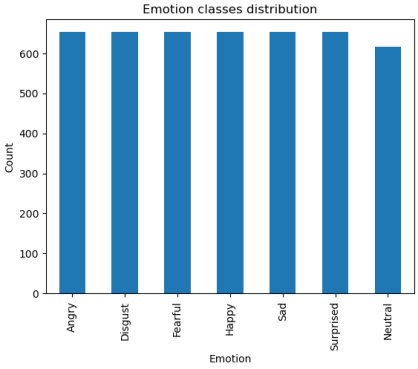



**Figure 2**. *Emotion class distribution*

Waveforms and spectrograms were used to illustrate differences in speech patterns across the different emotions (Figure 3). The waveform shows that the Angry sample had the highest peak amplitude (1.0), while the neutral sample showed the lowest peak amplitude (0.04). The spectrogram shows that the angry, fearful and happy emotions had strong frequencies in the lower frequency range, while the neutral, disgust and surprised had a more evenly distributed frequency profile.

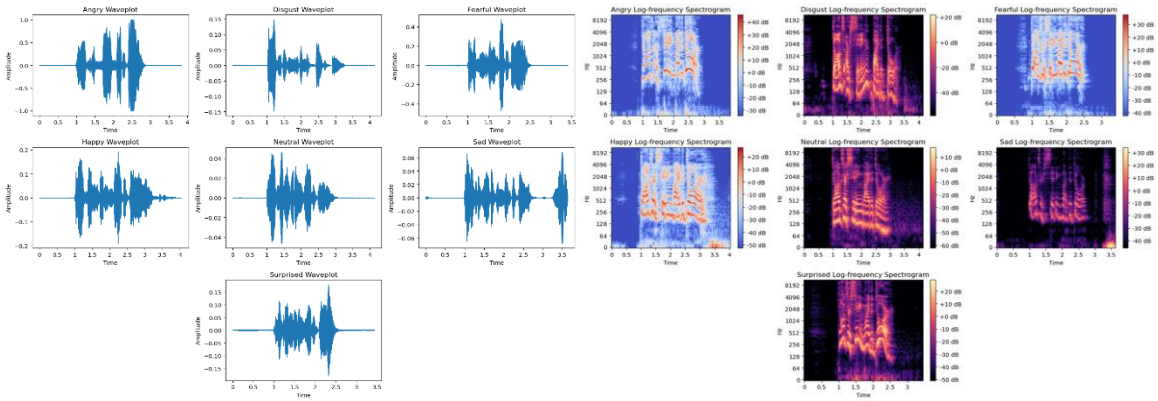



**Figure 3.** *Waveforms and spectrograms samples for each emotion*

*2.4 AI Techniques and tools*

**1.) Speech-to-text:**

- OpenAI's whisper-medium(English) pre-trained AI model to convert spoken speech into processed text

**2.) Emotion Prediction:**

- Training and testing with three supervised machine learning models:
    1.) Support Vector Machine (SVM)
    2.) Multi-Layer Perceptron (MLP)
    3.) Long Short-Term Memory (LSTM)

*2.5 LSTM Architecture*

LSTM models are a type of Recurrent Neural Networks (RNN) that are commonly used in SER problems due to their ability to model long-term dependencies (Makhmudov et al., 2024)**.** Studies have shown that multiple layers of LSTM can help improve the extraction of emotional information from speech (Yang et al., 2024). Additionally Bi-directional LSTM layers can improve extraction of emotional features by considering both the preceding and subsequent signals (Makhmudov et al., 2024).

Hence, the STE system utilizes a bi-directional, dual-layer LSTM with 256 hidden units to improve the predictive performance of the model**.** Dropout layers with a rate of 0.2 and batch normalization were added between the layers to improve the model's stability and reduce model overfitting. ADAM optimizer was used to adjust the learning rate during training with a starting learning rate of $1e^{-4}$. The model was trained over 100 epochs with batch size of 64. Early stopping was implemented to further reduce model overfitting.

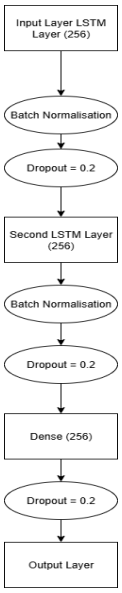



**Figure 4.** *LSTM Model Architecture*

*2.6 Project Architecture: Audio Data Pre-Processing*

The audio data and sample rate were obtained from the speech input using the Librosa library. Noise from the imported audio sample was reduced using the Noisereduce library. Reducing noise is an essential process as SER systems are highly sensitive to noise, resulting to reduced emotion prediction accuracies. Additionally, a Voice Activation Detection (VAD) filter was used during microphone recordings to filter out parts of recordings without speech, thereby reducing audio hallucinations. Speech from the cleaned data was transcribe into text using the Faster Whisper library.

Long audio data were divided into short audio clips (~2-5 seconds), with each clip processed individually. This was done due to two reasons. First, the ML models were trained exclusively on datasets consisting of short audio clips (Table 1). Second, emotion from a long speech tend to fluctuate over time. Hence, audio partitioning can help improve the predictive accuracy of the model in long, novel audio samples. This audio partitioning was also handled by the Faster Whisper library.

Features from each divided audio chunks were extracted using Mel frequency cepstral coefficients (MFCC). MFCC is a widely used feature extraction method for SER and other speech-related tasks (Likitha et al., 2017). MFCC extracts frequency-domain features from the audio data (Chatterjee et al., 2021). Based on these frequency-domain features it can analyse the speech textures and variations in speech frequency from audio samples (Gaddam, N. 2024). Hence, MFCC enables extraction and comparisons of emotional features between different emotions. A total of 100 features were extracted with the MFCC and were scaled using StandardScaler.

The dataset was split into 80% training and 20% test set. 5-fold cross validation was performed to reduce the risk of overfitting during tuning. SVM and MLP were tuned using the Grid Search CV and Randomized Search CV tools to further improve the models' performance

*2.7 Time frame*

The project was developed over the course of six weeks. The time frame of the creation of the project are as follows:

| | |
|---|---|
| Week 1 & 2 | • Project planning |
| Week 3 | • Implementing Speech-to-Text functionality |
| Week 4 | • Dataset Preparation<br>• Audio Processing and Feature Extraction |
| Week 5 | • Model Training, Testing and Tuning |
| Week 6 | • UI Implementation |

**Table 2.** *Time frame of the project creation*

# 3. Results and Evaluation

*3.1 Running Application*

A graphical user interface (GUI) was created using the Tkinter library that runs all the functionalities of the STE application. The interface provides the user with the option of either a microphone or an audio file as an input. The text window provides the user with the transcribed text and the predicted emotion by the LSTM model. Additionally, an interactable timestamp is available with the audio import option that allows the user to listen to the segmented audio clip. A summary of detected emotion is also presented, providing the user with information about the most frequently predicted emotion during the speech.

**Figure 5.** *Graphical User Interface of the STE Application*

### *3.2 Model Performance*

The LSTM model had the highest accuracies in predicting angry, fearful, neutral and sad emotions (Figure 6). However, the LSTM model performed poorly in predicting the disgust emotion. In contrast the SVM model had the highest accuracies in predicting disgust and happy emotions.

Furthermore, the LSTM model outperformed the SVM and MLP models, achieving an overall accuracy, recall, precision and F1-scores of 88% (Figure 7). However, the LSTM model had the slowest prediction time of 0.04 seconds compared to the SVM and MLP model. Overall, the LSTM model had the best prediction performance due to its ability to capture the long-term dependencies in speech data.

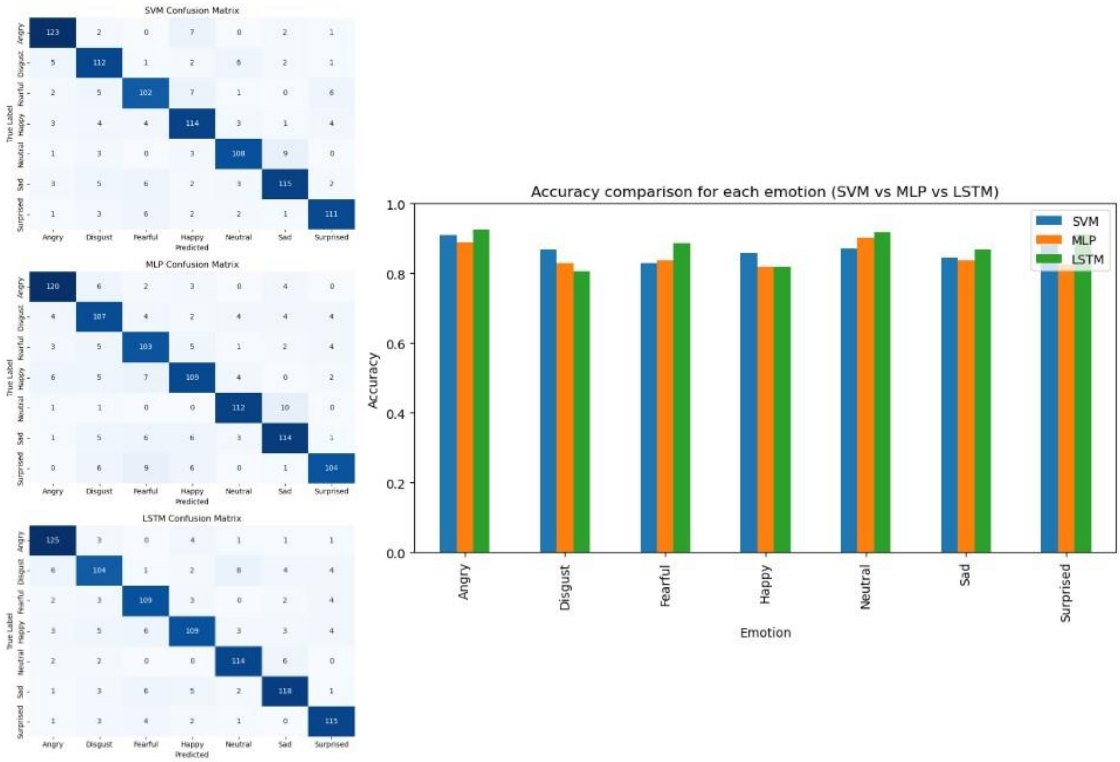
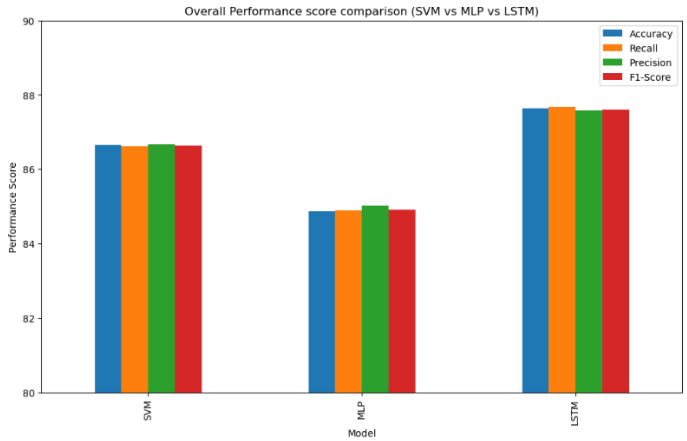



**Figure 6.** *Confusion matrix and accuracy score comparisons of SVM, MLP and LSTM models across all the emotion classes*




Performance Score Comparison

| | Accuracy | Recall | Precision | F1-Score | Prediction Time |
|---|---|---|---|---|---|
| **SVM** | 86.644592 | 86.617824 | 86.672129 | 86.626390 | 0.000506 |
| **MLP** | 84.878587 | 84.887878 | 85.026002 | 84.912870 | 0.000578 |
| **LSTM** | 87.637969 | 87.679365 | 87.589368 | 87.602452 | 0.040888 |

**Figure 7.** *Accuracy, Recall, Precision, F1-Score and Prediction time of SVM, MLP and LSTM models*

*3.3 Limitations*

One of the major limitations of the system is its inability to recognize and segregate multiple speakers at a time. Having multiple speakers can lead to poor results in emotion prediction and speech-to-text transcription because their voices are processed together as a single audio data. The second limitation is the poor prediction performance with the presence of high noise in the audio sample data. Although the system performs noise reduction techniques, extreme noise can distort the speech data and can therefore affect the extracted features by the MFCC. Thus, noise can reduce the accuracy of the model's predictions. Thirdly, the dataset samples were all recorded in English. Hence, the system can only successfully recognize emotions from English audio samples. Labelled audio samples from other languages are needed in the training dataset to expand the use case to other languages

*3.4 Difficulties/Challenges*

The ML models were initially trained and tested exclusively on the RAVDESS dataset. However, the models demonstrated poor predictive performance during testing evaluation (Figure 8). Due to the differences in speech patterns, such as accents, gender and age variation, training the models with a small and less diverse sample data can lead to inaccurate results causing model overfitting.

This problem was tackled by adding the SAVEE and TES datasets. This increased the training sample size and data variability. The SAVEE dataset improved speech variability in the training samples by having actors with a British accent, while the TES dataset included a young and an old actor. Therefore, the model's ability to generalize across diverse speech pattern has improved leading to a better predictive performance across the models

RAVDESS dataset

Performance Score Comparison

| | Accuracy | Recall | Precision | F1-Score | Prediction Time |
|---|---|---|---|---|---|
| SVM | 56.250000 | 55.406139 | 57.795502 | 55.631660 | 0.001001 |
| MLP | 61.805556 | 60.721939 | 62.708720 | 61.055294 | 0.000680 |
| LSTM | 57.291667 | 56.668967 | 57.090291 | 56.547556 | 0.040149 |

RAVDESS + T.E.S + SAVEE

Performance Score Comparison

| | Accuracy | Recall | Precision | F1-Score | Prediction Time |
|---|---|---|---|---|---|
| SVM | 86.644592 | 86.617824 | 86.672129 | 86.626390 | 0.000506 |
| MLP | 84.878587 | 84.887878 | 85.026002 | 84.912870 | 0.000578 |
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**Figure 8.** *Performance score comparison of models trained with RAVDESS dataset and RAVDESS & T.E.S & SAVEE datasets*

## 5. Discussion and Conclusions

The results have shown that the LSTM model is a fitting model to tackle this problem due to its ability to capture and detect temporal data**.** The performance results of the STE application highlighted the potential of the current AI technology in recognizing emotion from speech**.** However, due to the complex variation in speech data and the presence of noise in some audio samples, the system can inaccurately predict the emotion in a real-world audio scenario.

To improve the robustness of the SER systems in real-case scenarios, data augmentation can be used. The study by Makhmudov et al. (2024), used audio augmentation techniques such as pitch shifting, time stretching and Gaussian noise addition to increase the training sample and further improve the model's generalizability. Alternatively, emotion detection can be improved by integrating computer vision techniques. Changes in a person's facial expression can be used to detect the current emotional state of a person. This has been conducted in a study by Jaiswal et al. (2020), using a CNN model to detect emotion from faces.

Therefore, these enhancement techniques can be applied to the system to further improve the performance of STE in real-case scenarios.

## 6. References


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